

Set 8: Employee Performance Analysis

DSA – Data Handling and Visualization | Lab Experiment

R.No	Reg. No	Name
14	192524098	Divyasree Muthukumar

Aim

To analyse employee performance using a line chart, a department-wise bar chart and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
emp <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
write.csv(emp, "Employee_Performance.csv", row.names = FALSE) # for Tableau
```

Q1. Line Chart – Performance Trend over Time

Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

R Program:

```
# Q1. Line chart - performance score across years of service (time dimension)
emp_sorted <- emp[order(emp$Years_of_Service), ]
print(emp_sorted, row.names = FALSE)
cat("Correlation (Years of Service vs Score):",
    round(cor(emp$Years_of_Service, emp$Performance_Score), 3), "\n")

p_line <- ggplot(emp_sorted, aes(x = Years_of_Service, y = Performance_Score)) +
  geom_line(colour = "#0072B2", linewidth = 1.1) +
  geom_point(size = 4, colour = "#0072B2") +
  geom_text(aes(label = paste0("Emp ", Employee_ID, " (", Department, ")"),
    Performance_Score)), vjust = -1.2, size = 3.6) +
  scale_x_continuous(breaks = 2:8, limits = c(2, 8)) +
  scale_y_continuous(limits = c(70, 100)) +
  labs(title = "Employee Performance Trend over Years of Service",
    x = "Years of Service", y = "Performance Score") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_line)
ggsave("s8_q1_line.png", p_line, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
Employee_ID Department Years_of_Service Performance_Score
2          HR           3                92
```

1	Sales	5	85
3	Marketing	7	78

Correlation (Years of Service vs Score): -1

Output:

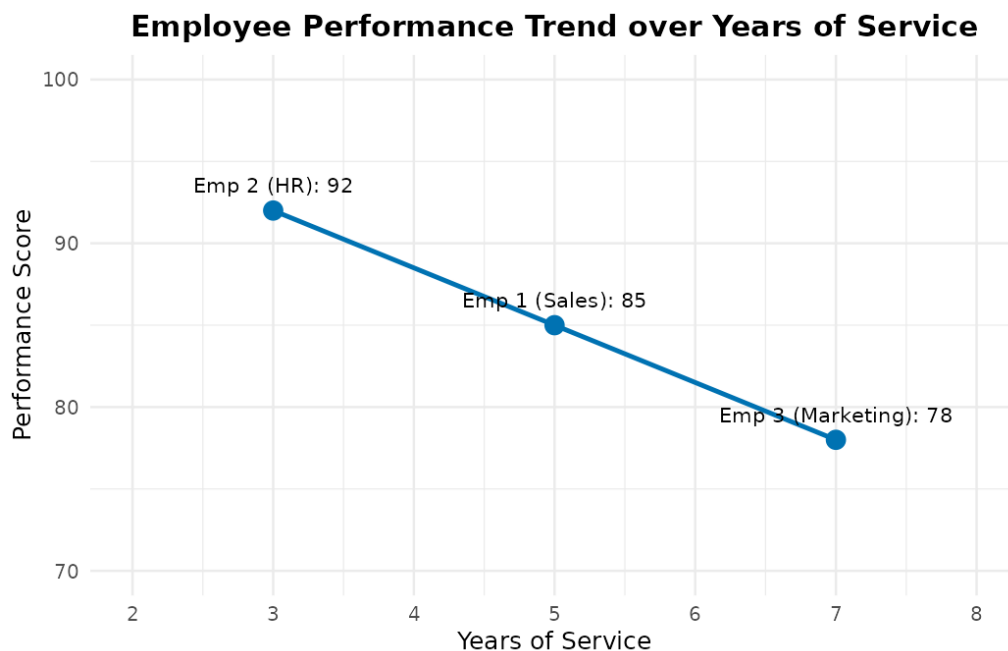


Figure 1: Performance score across years of service

Inference:

The dataset has no dates, so **Years of Service** is used as the time axis. Performance falls steadily as service increases: 92 (3 yrs, HR) → 85 (5 yrs, Sales) → 78 (7 yrs, Marketing), a perfect negative correlation ($r = -1$).

With only three employees, each from a different department, this is a cross-sectional comparison rather than a true trend for one person. More records over time would be needed to confirm it.

Q2. Bar Chart – Employees across Departments

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

R Program:

```
# Q2. Bar chart - number of employees per department
dept <- as.data.frame(table(emp$Department))
names(dept) <- c("Department", "Employees")
print(dept, row.names = FALSE)

p_bar <- ggplot(dept, aes(x = Department, y = Employees, fill = Department)) +
  geom_col(width = 0.55) +
  geom_text(aes(label = Employees), vjust = -0.5, size = 4.5) +
  scale_fill_manual(values = c(HR = "#009E73", Marketing = "#E69F00", Sales = "#0072B2")) +
  scale_y_continuous(limits = c(0, 1.5), breaks = c(0, 1)) +
  labs(title = "Distribution of Employees across Departments",
       x = "Department", y = "Number of Employees") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5), legend.position = "none")
print(p_bar)
ggsave("s8_q2_dept_bar.png", p_bar, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

Department	Employees
HR	1
Marketing	1
Sales	1

Output:

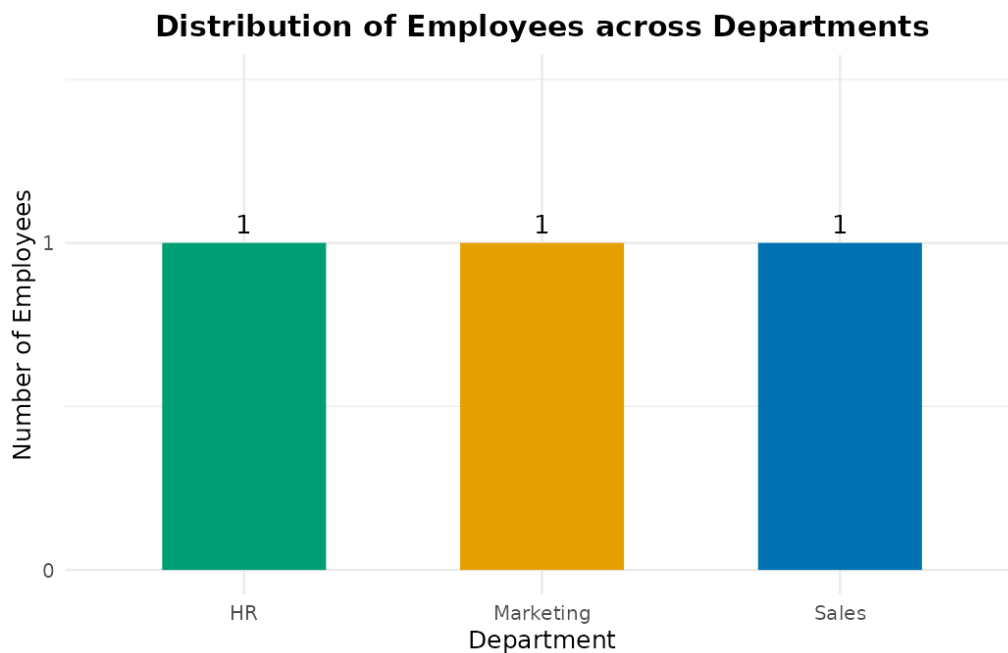


Figure 2: Number of employees per department

Inference:

Each department (HR, Marketing, Sales) has exactly one employee, so the workforce is evenly distributed across departments.

Q3. Table – Performance Data for Each Employee

Build a table to display the performance data for each employee. Label the table elements.

R Program:

```
# Q3. Labelled table of employee performance
tbl <- emp
names(tbl) <- c("Employee ID", "Department", "Years of Service", "Performance Score")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Employee Performance Data",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s8_q3_table.png", width = 1000, height = 330, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

```
Employee ID Department Years of Service Performance Score
1 Sales 5 85
2 HR 3 92
3 Marketing 7 78
```

Output:

Employee Performance Data

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Figure 3: Labelled employee performance table

Inference:

The table shows that Employee 2 (HR) has the highest score (92) with the fewest years of service (3), and Employee 3 (Marketing) has the lowest score (78) with the most service (7).

Q4. Tableau Dashboard – Interactive Exploration

Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

R Program:

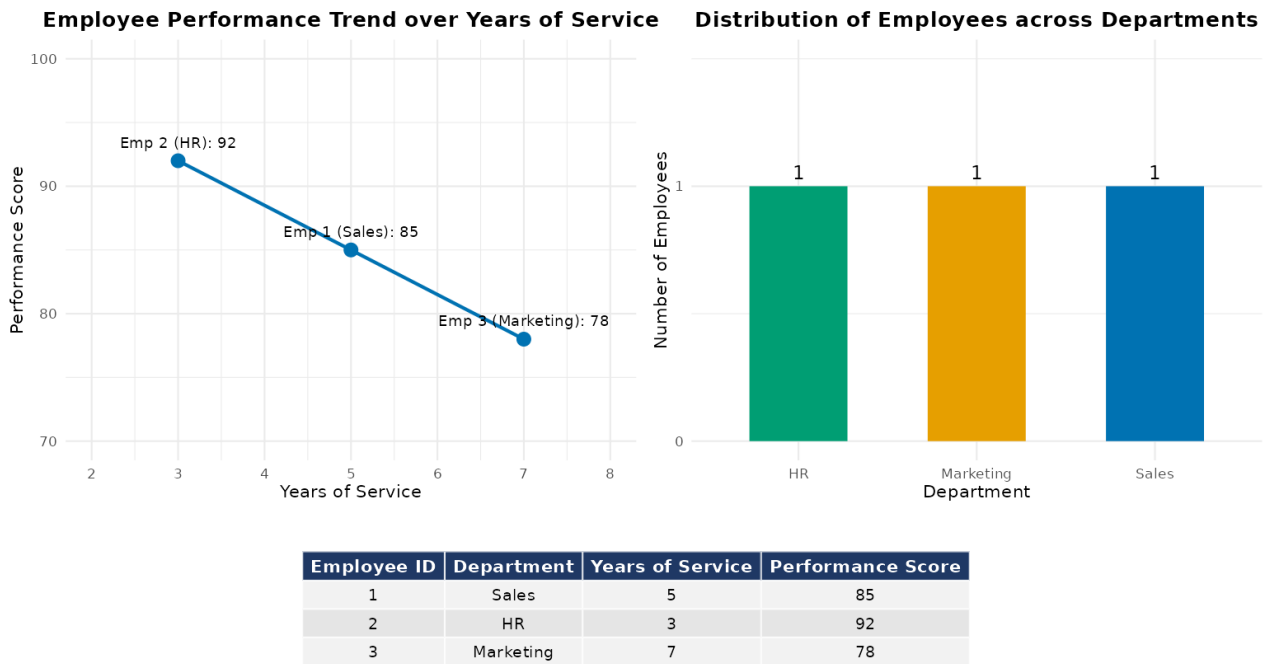
```
# Q4. Dashboard layout preview (Tableau dashboard is built from Employee_Performance.csv)
library(patchwork)
dash <- (p_line | p_bar) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Employee Performance Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s8_dashboard.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Employee_Performance.csv**. Convert **Employee ID** to a Dimension.
- 2 **Sheet 1 – "Performance Trend"**: drag Years of Service (as **Dimension, Continuous**) to Columns and AVG(Performance Score) to Rows. Set the Marks type to **Line** and turn on markers (Color → Markers). Drag Department and Employee ID to Label. Title the sheet and the axes.
- 3 **Sheet 2 – "Department Distribution"**: drag Department to Columns and **CNT(Employee ID)** to Rows. Drag Department to Color and show mark labels.
- 4 **Sheet 3 – "Performance Table"**: drag Employee ID, Department, Years of Service and Performance Score to Rows (discrete), or use Measure Names / Measure Values. Rename the headers.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: Performance Trend (top-left), Department Distribution (top-right), Performance Table (bottom). Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Employee_Performance) so one filter controls every view.
- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.
- 11 Add a **Department** quick filter (multiple values list) to the dashboard for selecting departments.

Dashboard Layout:

Employee Performance Dashboard



Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a department bar highlights that employee on the trend line and filters the table, which links each employee's department, tenure and score.

Result

The employee performance data was visualised with a line chart, bar chart and labelled table in R and combined into an interactive Tableau dashboard. In this sample, performance decreases as years of service increase (92 → 78), and employees are evenly spread across departments.